

STEFANO CASIRATI, Ph.D.

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RESEARCH INTERESTS

Eco-Hydrological Modeling, Computational Hydrology, Remote Sensing, Forests, Agriculture, Irrigation, Earth System Models, Food Energy Water Systems, Land-Atmosphere Interactions, Climate Teleconnections.

EDUCATION

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| 2023 | Ph.D. in Environmental Systems | University of California, Merced, California, USA
School of Engineering,
Advisors: Dr. Martha Conklin, Dr. Safeeq Khan |
| 2014 | M.Eng. Environmental Engineering, | Politecnico of Milan, Italy
Advisors: Dr. Maria Cristina Rulli, Dr. Paolo d'Odorico |
| 2011 | B.Eng. Environmental Engineering, | Politecnico of Milan, Italy
Advisor: Dr. Maria Cristina Rulli |

PROFESSIONAL EXPERIENCE

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| 2023 - Present | Postdoctoral Researcher - Environmental Science, Policy, and Management - University of California, Berkeley, USA |
| 2018-2023 | Graduate Student Researcher (GSR) and Teaching Assistant (TA) - Environmental Systems Graduate group - Department of Civil and Environmental Engineering - University of California, Merced, USA |
| 2015–2018 | Consultant, Environmental Engineer, and Energy Efficiency Analyst - Suntrading S.p.A. / Sunsaving S.r.l. / Solarfast S.p.A., Italy |
| 2015-2015 | Consultant, Environmental Engineer (HSE) - Tecnoambiente S.r.l., Italy |
| 2014–2015 | Research Intern “Virtual water analysis for food and energy” - Department of Civil and Environmental Engineering, Politecnico of Milan, Italy Research internship: |

PUBLICATIONS

Peer-Reviewed Journal Articles

- 2026 **Casirati, S.**, D’Odorico P., Fung I., Molod A., and Giroto M. (2026). “Differential impacts of irrigation on precipitation across the United States”. *Environmental Research Letters*, 21(15), 154028 <https://doi.org/10.1088/1748-9326/ae929d>
- Casirati, S.**, M. H. Conklin, and M. Safeeq. (2026). “Hydrological Response to Compounding Impacts of Climate Change and Forest Management in the Upper Kings River Basin, CA, USA.” *Ecohydrology* 19, no. 1: e70157. <https://doi.org/10.1002/eco.70157>
- 2025 **Casirati, S.**, D’Odorico, P., Reichle, R. H., Rouf, T., Stookey, M., & Giroto, M. (2025). A model-based evaluation of the effects of irrigation expansion on regional and global land surface climate. *Earth's Future*, 13, e2025EF006271. <https://doi.org/10.1029/2025EF006271>
- Casirati, S.**, Conklin, M., Nandi, S. and Safeeq, M. (2025), Effect of Forest Management Practices on Water Balance Across a Water–Energy Gradient in the Upper Kings River Basin, USA. *Ecohydrology*, 18: e2753. <https://doi.org/10.1002/eco.2753>
- 2023 **Casirati S.**, Conklin M.H. and Safeeq M. (2023) Influence of snowpack on forest water stress in the Sierra Nevada. *Front. For. Glob. Change* 6:1181819. <https://doi.org/10.3389/ffgc.2023.1181819>
- 2019 Rulli M.C., **Casirati S.**, Dell’Angelo J., Davis K.F., Passera C., D’Odorico P. (2019) Interdependencies and telecoupling of oil palm expansion at the expense of Indonesian rainforest. *Renewable and Sustainable Energy Reviews* 105, 499-512. <https://doi.org/10.1016/j.rser.2018.12.050>

Conference Proceedings

- 2015 D’Odorico P., Rulli M.C., **Casirati S.**, Davis K.F., Dell’Angelo J., Land use change induced by large scale land acquisitions. World Bank Conference on Land and Poverty, Washington DC, 23-27 March 2015.

PRESENTATIONS

- 2026 **Casirati S.**, D’Odorico P., Fung I., Giroto M., (2026) “Irrigation and Land–Atmosphere Interactions: Insights into Global Climate Patterns”, Oral Presentation, 106th AMS American Meteorological Society Annual Meeting, Houston, USA
- 2025 **Casirati S.**, D’Odorico P., Fung I., Giroto M., (2025) “Irrigation's Global and Regional Climate Impacts: Exploring Land–Atmosphere Feedbacks Across S2S Timescales”, Poster Presentation, Advancing Understanding of Land-Atmosphere Interactions and Processes on S2S Predictability Workshop, Boulder, Colorado, USA

- Safeeq, M. and **Casirati, S.**, (2025) “Forest Management in a Warming World: Enhancing Insights into Compounding Hydrologic Impacts”, Oral Presentation, European Geophysical Union General Assembly 2025, Vienna, Austria.
- 2024 **Casirati S.**, D’Odorico P., Giroto M., (2024) “Evaluating the effects of irrigation expansion on regional and global land surface climate using a Land Surface Model”, Poster Presentation, American Geophysical Union, Fall Meeting. San Francisco, California, USA
- Safeeq, M., **Casirati, S.**, Nandi, S. and Conklin, M., (2024) “Modeling the effects of forest thinning on the long-term water balance in a Mediterranean mixed-conifer basin”, Oral Presentation, American Geophysical Union, Fall Meeting. San Francisco, California, USA
- 2022 **Casirati S.**, Safeeq M., and Conklin M. H., (2022) “Predicting drought-related tree mortality in the Sierra Nevada.”, eLightning presentation Frontiers in Hydrology Meeting – San Juan, Puerto Rico, USA
- 2020 **Casirati S.**, Safeeq M., and Conklin M. H., (2020) “Influence of snow accumulation and ephemerality on recent tree mortality in the Sierra Nevada”, Oral Presentation, Yosemite Hydroclimate Meeting, Virtual.
- 2020 **Casirati S.**, Safeeq M., and Conklin M. H., (2020) “Influence of snow accumulation and ephemerality on recent tree mortality in the Sierra Nevada”, Poster Presentation, Sierra Nevada Research Institute - UC Merced Research Week 2020: Research Contributions: “Addressing Climate Change for the Region and the World” - University of California Merced, California, USA
- Casirati S.**, Safeeq M., and Conklin M. H., (2020) “Influence of snow accumulation and ephemerality on recent tree mortality in the Sierra Nevada”, Poster Presentation, Grad Visitation Poster Session: University of California Merced, California, USA
- 2019 **Casirati S.**, Safeeq M., and Conklin M. H., (2019) “Influence of snow accumulation and ephemerality on recent tree mortality in the Sierra Nevada”, Poster Presentation, American Geophysical Union, Fall Meeting. San Francisco, California, USA
- Casirati S.**, Safeeq M., and Conklin M. H., (2019) “Influence of snow accumulation and ephemerality on recent tree mortality in the Sierra Nevada”, Oral Presentation, Headwaters to Groundwater Annual Meeting, University of California Davis, California, USA
- Casirati S.**, Safeeq M., and Conklin M. H., (2019) “Influence of snow accumulation and ephemerality on recent tree mortality in the Sierra Nevada”, Oral Presentation, Symposium: "Climate change and the ecology of Sierra Nevada forests", University of California Merced, California, USA

COMPETITIVE SCIENTIFIC PROPOSALS

- (Pending) NASA Early Career Investigator Program in Earth Science: Hydroclimatic Feedbacks and the Global Food-Water Nexus: Balancing Yield Targets with Sustainable Irrigation (Role: PI)

AWARDS and FELLOWSHIPS

2023	UC Merced ES Summer Fellowship (\$7500)
2022	AGU FIHM Travel Grant (\$1000)
2022	UC Merced ES Summer Fellowship (\$7000)
2019	Fall-1819 UC Merced ES Bobcat Travel Fellowship (\$1500)

TEACHING

University of California, Merced

ENVE 181: Field Methods in Snow Hydrology - Spring 2022 and Spring 2023 (*Teaching Assistant*)

Responsibilities: Supported field activities involving in situ measurements of snowpack properties. Assisted students with field methods, data collection, and the practical application of snow hydrology concepts.

CE 010: Surveying and Geomatics - Fall 2022 (*Teaching Assistant*)

Responsibilities: Designed and conducted fieldwork and lab experiments using a range of surveying and geomatics equipment including total stations, GNSS RTK, and laser scanning systems. Organized and adapted field activities to accommodate different class sizes across the two lab sections. Prepared and demonstrated equipment before each lab and provided brief introductions to the underlying theory, objectives, and field procedures. Structured activities to promote group work, discussion, and active student engagement.

MATH 011: Calculus I - MATH 011 - Spring 2022 (*Teaching Assistant*)

Responsibilities: Conducted laboratory exercises for Calculus I, guiding students through integration and differentiation concepts. Assisted students with exercises, reviewed and corrected worksheets, and adapted instruction to accommodate different learning paces and levels of understanding. Developed brief instructional explanations and examples to address common challenges and reinforce course concepts.

ENGR 180: Spatial Analysis & Modeling - Fall 2020 (*Guest Lecture*)

MENTORING

2025-	Chloe Faehndrich (UC Berkeley, Ph.D.)
2024-	Sarah Ryu (UC Berkeley, BS)
2020	Breanna Xiong (UC Davis, BS <i>Summer Intern California Ecosystem Climate Solutions project</i>)

ACADEMIC and PUBLIC SERVICE

2025	Panelist, NASA Grant Proposal Reviewer. Future Investigators in NASA Earth and Space Science and Technology (FINESST) Earth Panelist (Spring 2025)
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2022 Equipment removal for the Hemlock Forest Restoration Project, Sierra Nevada Research Institute

2021 Environmental Systems seminar graduate student lead

2020 Environmental Systems seminar graduate student lead

2019 Weir repair activities at the Kings River Experimental Watershed, US Forest Service

Reviewer: *Journal of Hydrology, Journal of Advances in Modeling Earth Systems, Journal of Hydrometeorology, Theoretical and Applied Climatology, Review of Geophysics, Water Resources Research*

ADDITIONAL TRAINING

2021 Principle of Pedagogy course

2020 Processed-Based Hydrological modeling – University of Saskatchewan - Centre for Hydrology

2019 CUAHSI Master Class: Advanced Techniques in Watershed Science - University of Arizona - Biosphere 2

PROGRAMMING AND COMPUTING SKILLS

- **Operating systems:** Windows, Linux, Unix
- **Programming Languages:** R, Python, Fortran
- **Other computer tools:** ArcGIS, QGIS, CropWat, SWAT+, SUMMA, Google Earth Engine, NASA GEOS, NASA CLSM, LaTeX, Microsoft Office

LANGUAGES

- Fluent in Italian and English